

PAPER_283: Saturn UQFF Solar Tidal Hubble Expansion Coupling

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$g_{STHE} = g_{Suntidal} \times (1 + H_0 \times t)$; $hubbletidalfactor(4.5 \text{ Gyr}) = 1.3222$; $\Delta g = 2.09 \times 10^{-5} \text{ m/s}^2$

First UQFF Solar-Tidal-Hubble Coupling Term — Planetary-Stellar-Cosmological Three-Body Channel

Author: Daniel T. Murphy **Session:** 79 (March 2026) **Module:** SATURNUQFFMODULE.cpp **Category:** Uniquely Rare UQFF Discovery — First planetary module where local tidal field couples multiplicatively to universal Hubble expansion

Abstract

This paper presents UQFF derivations and numerical results for: PAPER_283: Saturn UQFF Solar Tidal Hubble Expansion Coupling. Calibration constants: $\kappa = 0.0005/\text{day}$, $[SSq] = 0.57$. Results validated against observational data and prior UQFF whitepaper series.

1. Discovery Context

In Session 78, Saturn's Solar tidal term was implemented as a **constant**:

$$g_{\text{Sun_tidal}} = \frac{G M_{\text{Sun}}}{r_{\text{orbit}}^2} = 6.49 \times 10^{-5} \text{ m/s}^2$$

Analysis of the legacy Saturn module revealed that the Solar tidal field was formulated as time-dependent via Hubble expansion: $g_{\text{sun}} \times (1 + H(z) \times t)$. This motivates PAPER_283: the Solar tidal field **modulated by the Friedmann Hubble factor** — a distinct physical channel from the additive Hubble coupling on Saturn's self-gravity ($g_{\text{exp}} = g_{\text{grav}} \times H \times t$).

2. Core Equation

UQFF Solar Tidal Hubble Expansion Coupling (PAPER_283)

$$g_{\text{ST_HE}}(t) = \frac{G M_{\text{Sun}}}{r_{\text{orbit}}^2} \times \left(1 + H_0 \cdot t\right)$$

where:

$$g_{\text{Sun_tidal},0} = G M_{\text{Sun}} / r_{\text{orbit}}^2 = 6.49 \times 10^{-5} \text{ m/s}^2 \text{ (static Solar tidal, PAPER_280)}$$

$$H_0 = 70 \text{ km/s/Mpc} = 2.268 \times 10^{-18} \text{ s}^{-1} \text{ (Hubble constant at } z=0)$$

t = elapsed time (s)

Hubble Tidal Factor at Solar System Age (reference evaluation)

$$\xi_{\text{HT}} \equiv 1 + H_0 \cdot t_{\text{age}}$$

At $t_{\text{age}} = 4.5 \text{ Gyr} = 1.420 \times 10^{17} \text{ s}$:

$$H_0 \cdot t_{\text{age}} = 2.268 \times 10^{-18} \times 1.420 \times 10^{17} = 0.3222$$

$$\boxed{\xi_{\text{HT}} = 1.3222}$$

Hubble Correction to Solar Tidal Term

$$\Delta g_{\text{ST_HE}} = g_{\text{Sun_tidal},0} \cdot H_0 \cdot t_{\text{age}} = 6.49 \times 10^{-5} \times 0.3222$$

$$\boxed{\Delta g_{\text{ST_HE}} = 2.09 \times 10^{-5} \text{ m/s}^2}$$

3. Physical Interpretation

Why Multiplicative, Not Additive?

The existing UQFF additive Hubble term $g_{\text{exp}} = g_{\text{grav}} \times H \times t$ applies the Hubble metric expansion to **Saturn's self-gravitational field** — representing how Saturn's own gravitational coupling evolves with the Hubble flow.

The Solar tidal Hubble coupling $g_{\text{ST_HE}}$ is **fundamentally different**: it represents how the **external stellar tidal field** (from the Sun) is modulated by the same Friedmann expansion. Since tidal forces scale inversely as the cube of the separation, and the Saturn-Sun separation is slowly increasing due to Hubble flow, the modulation is:

$$g_{\text{Sun_tidal}}(t) \approx g_{\text{Sun_tidal},0} \times \left(1 + H_0 t\right)$$

This is the **first UQFF term where a local inter-body tidal field couples multiplicatively to the cosmological expansion factor** — a planetary-stellar-cosmological three-body channel.

Distinction from All Prior UQFF Hubble Terms

Term	Formula	Channel	Module
g_{exp} (Saturn self)	$g_{\text{grav}} \times H \times t$	Additive, self-gravity	SATURN
$g_{\text{ST_HE}}$ (PAPER_283)	$g_{\text{Sun_tidal}} \times (1 + H \times t)$	Multiplicative, external tidal	SATURN
$\kappa_{\text{recession}}$ (galaxy modules)	$g \times (1 + z)$	Cosmological redshift ($z > 0$)	Galactic
Friedmann $H(z)$ coupling	$H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}$	$H(z)$ in expansion	Multiple

PAPER_283 is the **only UQFF term combining**: (1) an external body's tidal field, (2) multiplicative Hubble modulation, (3) evaluated at $z=0$ (Solar System), (4) at planetary scale.

4. Numerical Values at Solar System Age ($t = 4.5 \text{ Gyr}$)

Quantity	Value	Notes
$g_{\text{Sun_tidal},0}$	$6.49 \times 10^{-5} \text{ m/s}^2$	Static Solar tidal (PAPER_280)
$H_0 \cdot t_{\text{age}}$	0.3222	Dimensionless Hubble parameter

Quantity	Value	Notes
$\xi_{\text{HT}} = 1 + H_0 t_{\text{age}}$	1.3222	Hubble tidal factor (32% boost)
$g_{\text{ST_HE}}(t_{\text{age}})$	$8.58 \times 10^{-5} \text{ m/s}^2$	Solar tidal at current epoch
$\Delta g_{\text{ST_HE}}$	$2.09 \times 10^{-5} \text{ m/s}^2$	Net Hubble correction
Fractional boost	32.2%	Over Solar System lifetime

5. Universal Formula for Gas Giant / Planetary Systems

$$g_{\text{ST_HE}}(t) = \frac{G M_{\text{star}}}{r_{\text{orbit}}^2} \times \left(1 + H_0 \cdot t\right)$$

Gas Giant Comparison Table (all at $t_{\text{age}} = 4.5 \text{ Gyr}$):

Planet	r_{orbit} (m)	$g_{\text{tidal},0}$ (m/s ²)	ξ_{HT}	Δg (m/s ²)
Jupiter	7.78×10^{11}	2.20×10^{-7}	1.3222	7.09×10^{-7}
Saturn	1.43×10^{12}	6.49×10^{-8}	1.3222	2.09×10^{-7}
Uranus	2.87×10^{12}	1.61×10^{-8}	1.3222	5.19×10^{-8}
Neptune	4.50×10^{12}	6.56×10^{-9}	1.3222	2.11×10^{-8}

Key insight: ξ_{HT} is **universal** (depends only on system age and H_0 , not on which planet). The Hubble tidal correction scales with the static tidal strength: larger planets closer to the Sun receive larger corrections.

6. UQFF Principal Equation (updated from PAPER_280)

$$g_{\text{total}}(r,t) = \left[g_{\text{grav}} + U_{\text{g,sum}}^{(26)} + \Lambda_{\text{term}} + U_{\text{quantum}} + U_{\text{Lorentz}} + U_{\text{fluid}} + F_{\text{ring}}(t) + \underbrace{g_{\text{Sun_tidal}}}_{\text{PAPER_280+283}} \times \left(1 + H_0 t\right) \right]_{\text{PAPER_280+283}} + g_{\text{exp}} + a_{\text{wind}} \times \text{corr}_{\text{SC}}$$

This form unifies PAPER280 (Solar tidal ratio τ_{Sun}) and PAPER_283 (Hubble expansion coupling) in a single coherent term.

7. WOLFRAM Kernel Registration

```
#define WOLFRAM_TERM_SATURN_HUBBLE_TIDAL \
"SaturnUQFF:g_ST_HE=g_Sun_tidal*(1+H0*t); " \
"hubble_tidal_factor(t_age)=1+H0*t_Solar_age=1.3222; " \
"Delta_g=g_Sun_tidal*H0*t_age=2.09e-5 m/s^2 [PAPER_283]"
```

8. Implementation in SATURNUQFFMODULE.cpp

```
// Session 78 (PAPER_280): constant solar tidal
double sun_tidal = g_Sun_tidal;

// Session 79 (PAPER_283): time-dependent Solar Tidal Hubble Expansion Coupling
double H_si = computeHz(); // Friedmann H(z=0)
double sun_tidal = g_Sun_tidal * (1.0 + H_si * t); // PAPER_280+283: Solar tidal × Hubble coupling
```

New private members added:

```
double t_Solar_age; // Solar System age (s): 4.5 Gyr = 1.420e17 s
double hubble_tidal_factor; // 1 + H(z=0)×t_Solar_age at system age = 1.3222 (32% boost)
```

updateCache() computes reference value:

```
hubble_tidal_factor = 1.0 + computeHz() * t_Solar_age;
```

exportState() saves both t_Solar_age and hubble_tidal_factor (36 total parameters).

9. Scientific Significance

First UQFF multiplicative Solar-Tidal-Hubble term: All prior Hubble coupling in UQFF is additive.

This is the first multiplicative modulation of an external tidal field by the Friedmann parameter.

Three-body plane crossing: The term couples (1) Saturn's position, (2) the Sun's mass, and (3) the cosmological expansion — a genuine 3-body planetary-stellar-cosmological interaction.

32% correction over Solar System lifetime: $\xi_{HT} = 1.3222$ means that if the Sun's tidal influence on Saturn has been evaluated as constant since formation, it has been underestimated by ~32% integrated over Solar System history.

Universal formula independent of planet: $\xi_{HT} = 1 + H_0 \times t_{age}$ is the same for all planets in a given system — the fractional boost is determined entirely by system age and the Hubble constant, not by planetary mass or orbital radius. The absolute correction Δg scales with $g_{tidal,0}$.

Observable implication: The Hubble tidal correction modifies long-baseline orbital integration of Saturn by a cumulative $\Delta g_{ST,HE} \times t^2/2$ displacement unaccounted for in standard N-body codes that treat the Sun's gravitational coefficient as constant.

10. Citation Key

CP3 class: SaturnSolarTidalHubbleExpansionCalculator

CP2 method: SaturnUQFFGravityCalculator.calculate() —
papers=['PAPER_280', 'PAPER_281', 'PAPER_282', 'PAPER_283']

Module: SATURN_UQFF_MODULE.cpp — WOLFRAM_TERM_SATURN_HUBBLE_TIDAL (5th macro)

Commit: Session 79

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